# Day 5 Student Handout

Geometry Summer Credit | Informal Triangle Congruence

| Name: | Date: | Period: |
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**Learning Targets**

I can use SSS, SAS, ASA, and AAS as triangle congruence shortcuts. I can use congruent triangles to find missing side lengths and angle measures.

## Lesson 5A: Triangle Congruence Shortcuts

### Warm-Up: Bring Back the Week

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| 1. | Find the missing angle in a triangle with angles 47 degrees and 68 degrees. |

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| 2. | Triangle ABC is congruent to triangle DEF. If AB = 9 and angle C = 41 degrees, what parts in triangle DEF match those? |

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| 3. | Reflect P(-4, 7) across the y-axis, then rotate Q(3, -5) 180 degrees about the origin. |

### Concept Launch

**How Much Information Is Enough?**

Triangle congruence shortcuts tell us when two triangles must be congruent without measuring every side and every angle. Today we use them informally, as reasoning tools, not as formal proofs.

| Problem | Work / notes |
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| SSS: Two triangles have side lengths 5, 7, 9 and 5, 7, 9. Are they congruent? | Enough information?<br>Shortcut:<br>Conclusion: |
| SAS: Two sides and the included angle match: 6, 80 degrees, 10. Are the triangles congruent? | Included angle means:<br>Shortcut:<br>Conclusion: |

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| ASA: Two angles and the included side match: 42 degrees, side 12, 71 degrees. | Included side means:<br>Shortcut:<br>Conclusion: |
| AAS: Two angles and a non-included side match: 38 degrees, 64 degrees, side 9. | Non-included side means:<br>Shortcut:<br>Conclusion: |

## Important Non-Examples

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| 4. | SSA is not a reliable shortcut. Sketch or explain why two sides and a non-included angle might not force one unique triangle. |

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| 5. | AAA shows triangles are similar, but not necessarily congruent. Explain the difference. |

# Day 5 Student Handout

Lesson 5B | Using Congruent Triangles

## Lesson 5B: Corresponding Parts After Congruence

### The Move After the Shortcut

Once triangles are congruent, matching sides are equal and matching angles are equal. The triangle order matters: ABC congruent to XYZ means A matches X, B matches Y, and C matches Z.

| Problem | Work / notes |
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| Triangle ABC is congruent to triangle XYZ. If AB = 14, BC = 11, and angle C = 56 degrees, find XY, YZ, and angle Z. | Corresponding order:<br>Answers: |
| Triangles MNO and PQR are congruent by SAS. MN = 8, angle N = 73 degrees, and NO = 13. What matches those parts? | MNO -> PQR<br>Matching parts: |
| Given triangle DEF congruent to triangle JKL, find x if DE = 2x + 3 and JK = 17. | Equation:<br>Solve:<br>Side length: |
| Given triangle RST congruent to triangle UVW, find x if angle S = 5x + 10 and angle V = 80 degrees. | Equation:<br>Solve:<br>Angle measure: |

### Practice

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| 6. | Name the shortcut: three pairs of matching sides are congruent. |

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| 7. | Name the shortcut: two angles and the side between them are congruent. |

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| 8. | Triangle ABC is congruent to triangle DEF. If AC = 15, what side in triangle DEF is also 15? |

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| 9. | Find x: triangle PQR congruent to triangle LMN, angle Q = 3x + 8, and angle M = 65 degrees. |